

Forecasting Monthly Median Dwelling Prices in Porto: Smoothing, Decomposition and ARIMA Methods

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FMTS — Time Series Forecasting
Student numbers: *[fill in]*

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Abstract

This report forecasts monthly median dwelling prices in Porto (EUR/m²) using multiplicative smoothing and decomposition methods, together with an ARIMA model fitted on a log-transformed, differenced series. The sample spans January 2011–February 2026 ($n = 182$). A hold-out test set of 24 months (March 2024–February 2026) is used to compare methods. ETS achieves the lowest test RMSE (326 EUR/m²); all methods under-forecast the accelerated growth in the test period. Out-of-sample 95% prediction intervals for the next 24 months are reported for each approach.

Contents

1	Introduction and data description	3
1.1	Objective	3
1.2	Data source	3
1.3	Features of the series	3
1.4	Train/test split	4
2	Decomposition methods	5
2.1	Multiplicative decomposition	5
3	Smoothing methods	7
3.1	Holt-Winters (multiplicative)	7
3.2	ETS and STLF	7
4	ARIMA modelling	8
4.1	Transformations and differencing	8
4.2	Model short-list and AIC	10
4.3	Diagnostic checking	10
5	Forecast comparison	12
6	Out-of-sample forecasts	12
7	Benefits and limitations	13
8	Software and reproducibility	13
	References	13
	Declaration of Authenticity	13

1 Introduction and data description

1.1 Objective

The objective is to forecast a real monthly time series using smoothing, decomposition, and ARIMA/SARIMA methods, to compare their predictive performance on a test set, and to produce out-of-sample forecasts with 95% prediction intervals.

1.2 Data source

The variable is the monthly median price per square metre (EUR/m²) for dwellings in Porto (series code 11A1312: Porto). Data are stored in `timeSeriesPorto.csv` and cover January 2011–February 2026.

1.3 Features of the series

Figure 1 shows a long upward trend: prices remain near 1 000 EUR/m² until about 2015, then increase steadily with faster growth after 2020. A mild seasonal pattern is visible; the seasonal boxplot (Figure 2) confirms that within-year variation is small relative to the trend. A spike around 2022–2023 is followed by a correction and renewed growth.

The ACF and PACF of the levels (Figure 3) decay slowly with spikes at lags 12, 24 and 36, indicating non-stationarity and annual seasonality. Following the course guidelines, unit-root tests designed for non-seasonal series (ADF, PP, KPSS) are not applied; stationarity is assessed through visual inspection, ACF/PACF, and differencing.

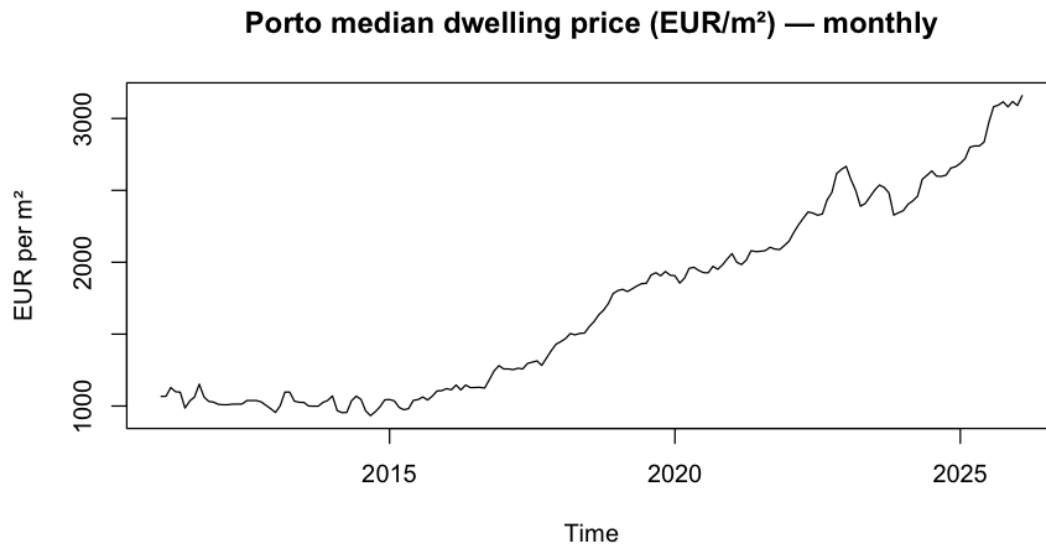


Figure 1: Monthly median dwelling price in Porto (EUR/m²).

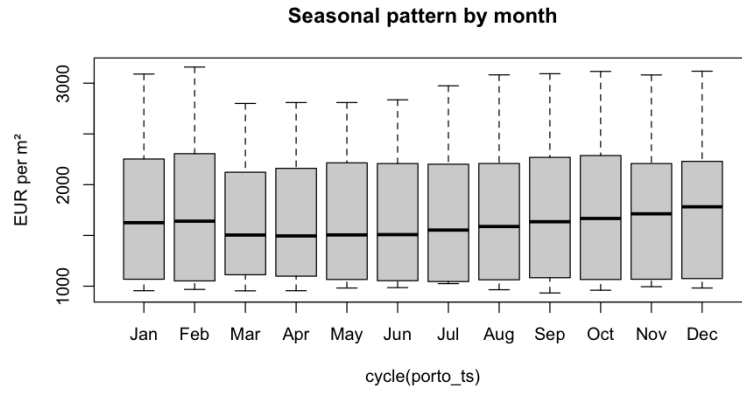


Figure 2: Seasonal boxplot by calendar month.

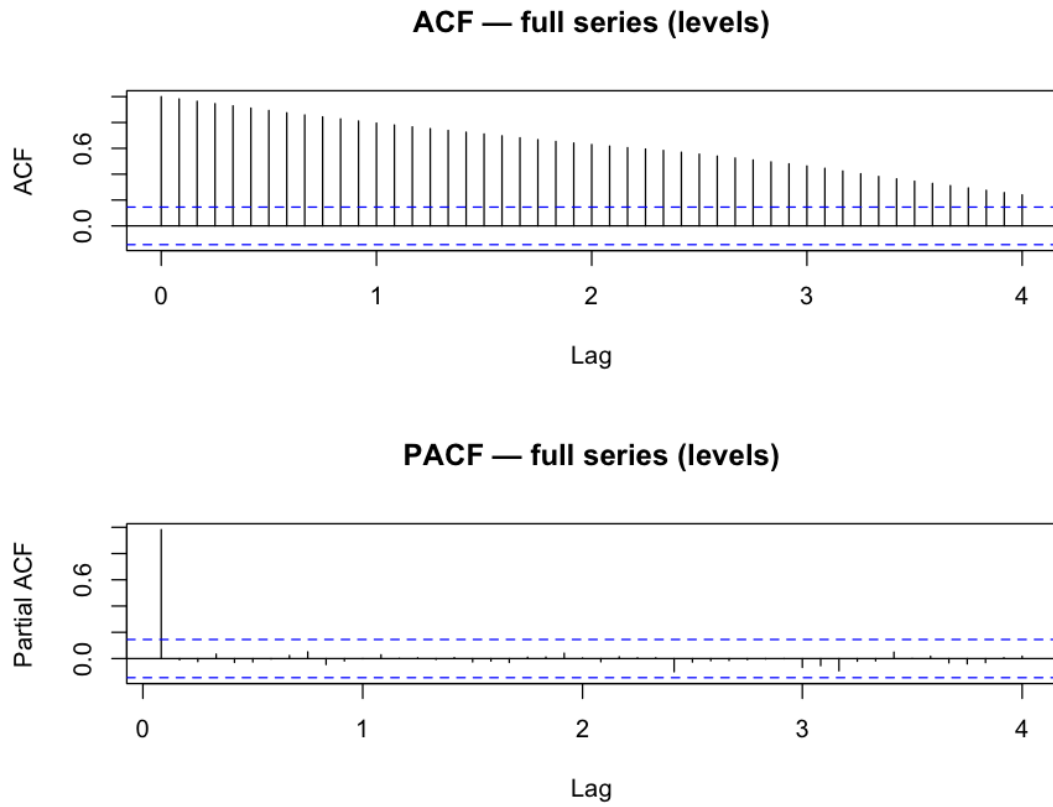


Figure 3: ACF and PACF of the series in levels.

1.4 Train/test split

The last $h = 24$ months (March 2024–February 2026) form the test set. All models are estimated on the training sample (January 2011–February 2024).

2 Decomposition methods

2.1 Multiplicative decomposition

Because seasonal effects are more naturally expressed as proportional to the level when a strong trend is present, **multiplicative** decomposition is used:

$$y_t = T_t \times S_t \times R_t,$$

where T_t is trend-cycle, S_t seasonal, and R_t remainder.

Figure 4 shows the classical multiplicative decomposition. Seasonal factors fluctuate around 1.0 with a narrow range (roughly 0.98–1.05), confirming that seasonality is small in relative terms but non-negligible in theory.

STL is applied to $\log(y_t)$ so that additive components on the log scale correspond to multiplicative structure on the original scale (Figure 5). The seasonally adjusted series is obtained as $y_t^{\text{SA}} = y_t / S_t$ (Figure 6).

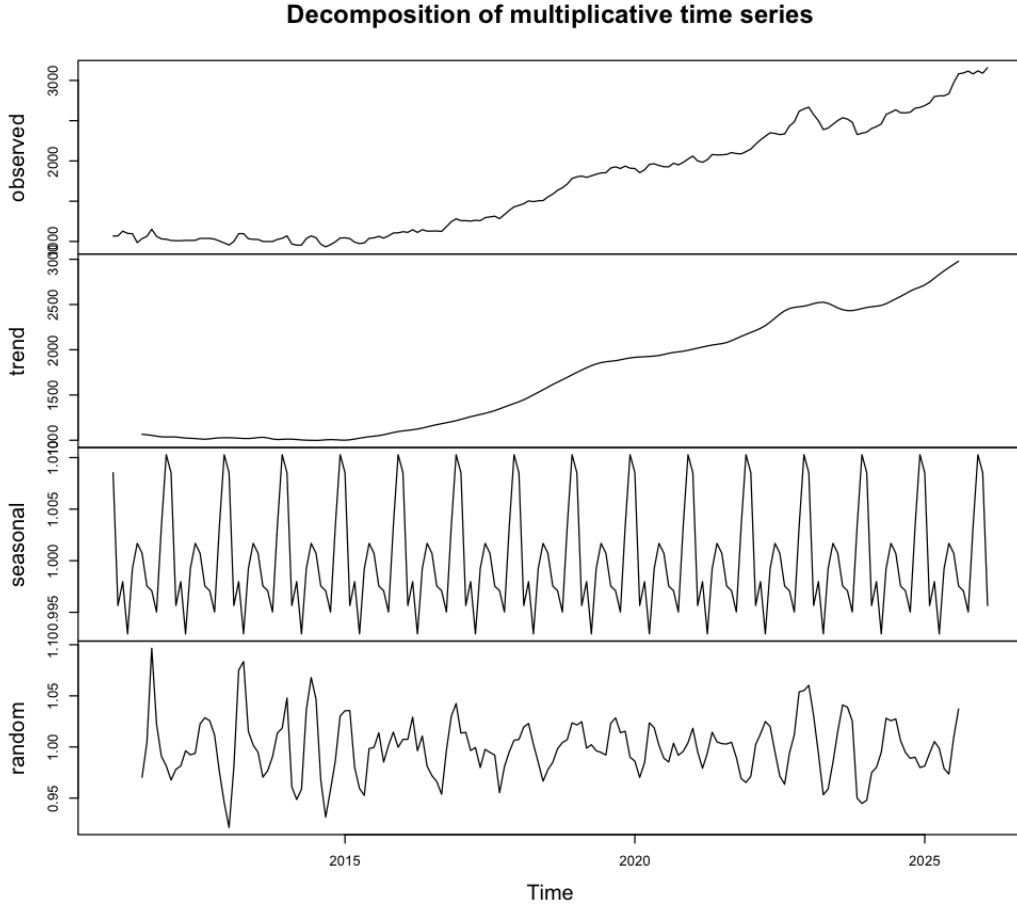


Figure 4: Classical multiplicative decomposition.

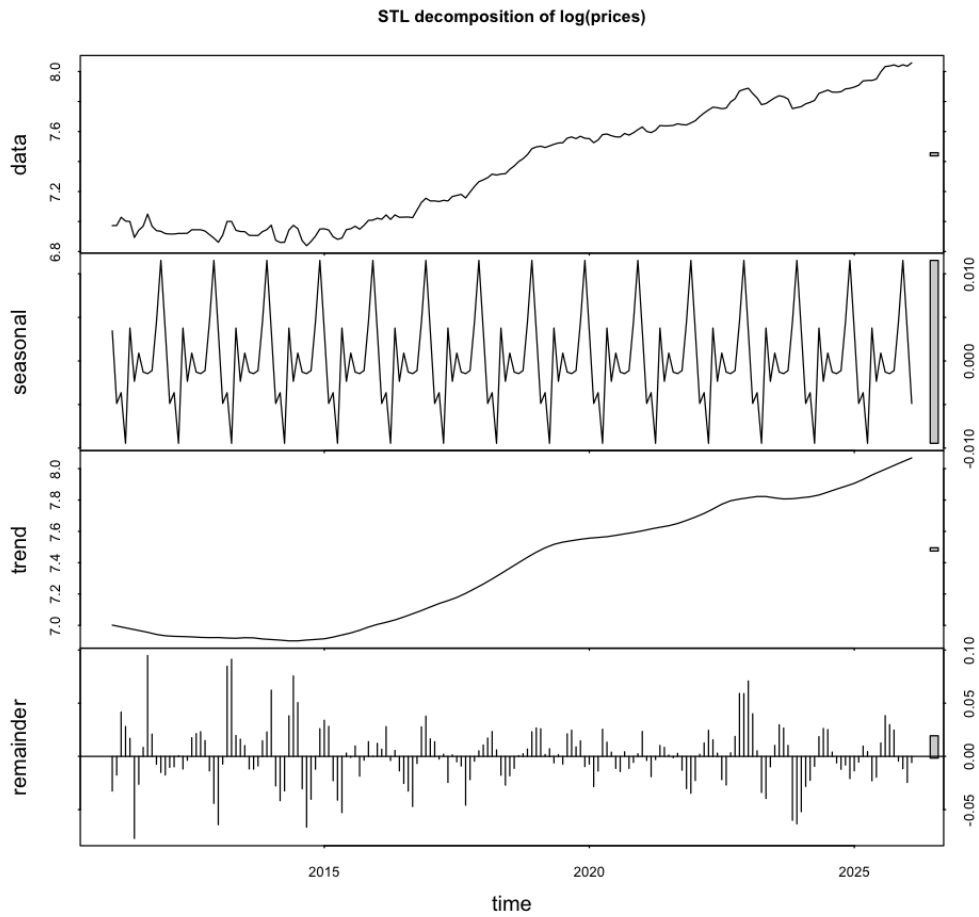


Figure 5: STL decomposition of $\log(y_t)$.

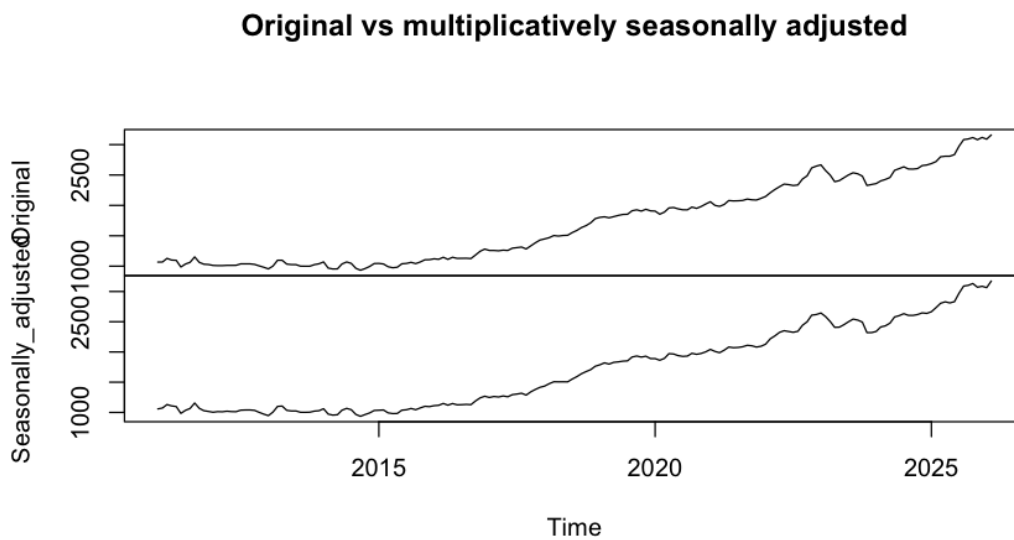


Figure 6: Original vs. multiplicatively seasonally adjusted series.

3 Smoothing methods

3.1 Holt-Winters (multiplicative)

Holt-Winters with **multiplicative seasonality** is fitted to the training sample. Estimated smoothing parameters are $\alpha = 0.804$, $\beta = 0.018$, and $\gamma = 1.000$. Seasonal factors s_m range from about 0.983 (February) to 1.047 (October). A damped-trend variant is also considered.

3.2 ETS and STL

The `ets()` function selects **ETS(A,A,N)** on the training window (additive trend, no seasonal term; AIC = 1981.5). STL combines STL with ETS forecasting on the seasonally adjusted component.

Figures 7–9 show each method against the test set with 95% prediction intervals.

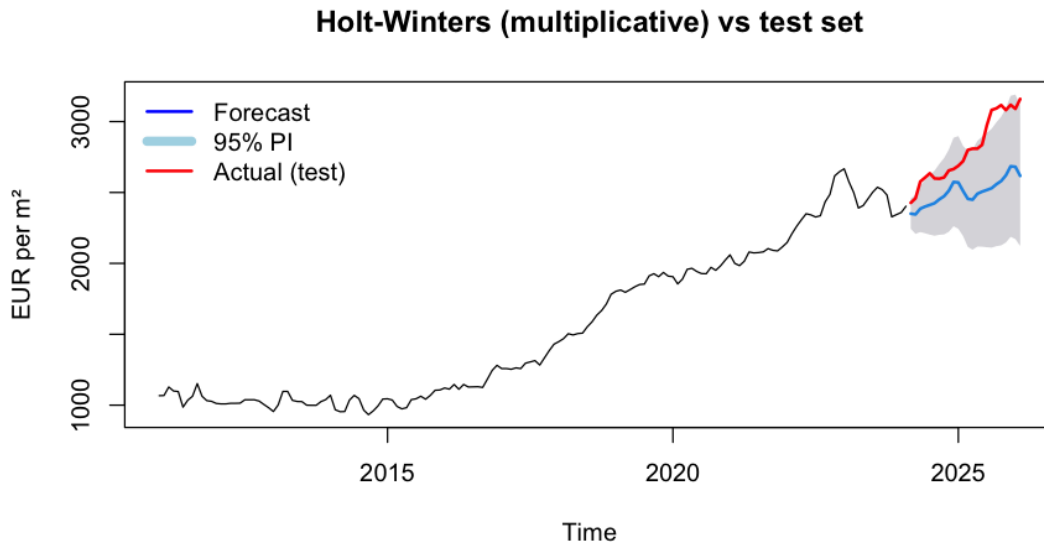


Figure 7: Holt-Winters (multiplicative) vs. test set.

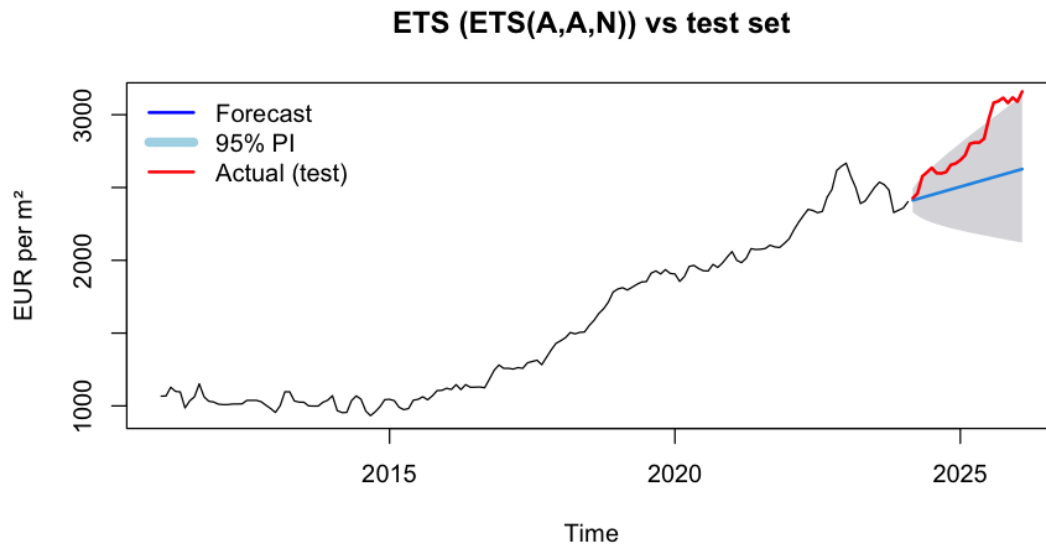


Figure 8: ETS vs. test set.

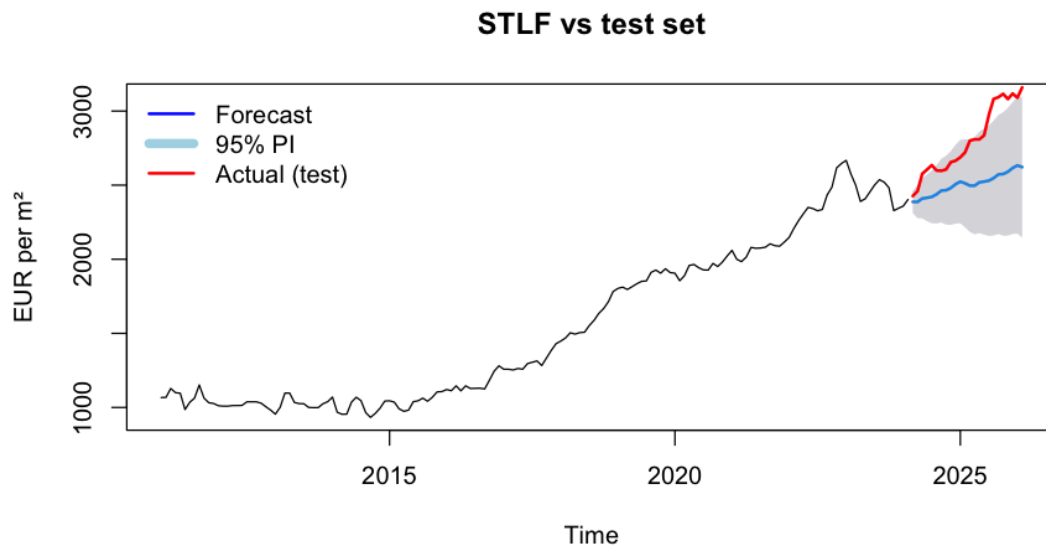


Figure 9: STL vs. test set.

4 ARIMA modelling

4.1 Transformations and differencing

Prices grow over time and variance increases with the level. A **log transformation** stabilises variance; **first differencing** of $\log(y_t)$ removes the trend and yields a series that is approximately stationary (Figures 12–10). The ACF/PACF of the differenced log series (Figure 11) guide model identification.

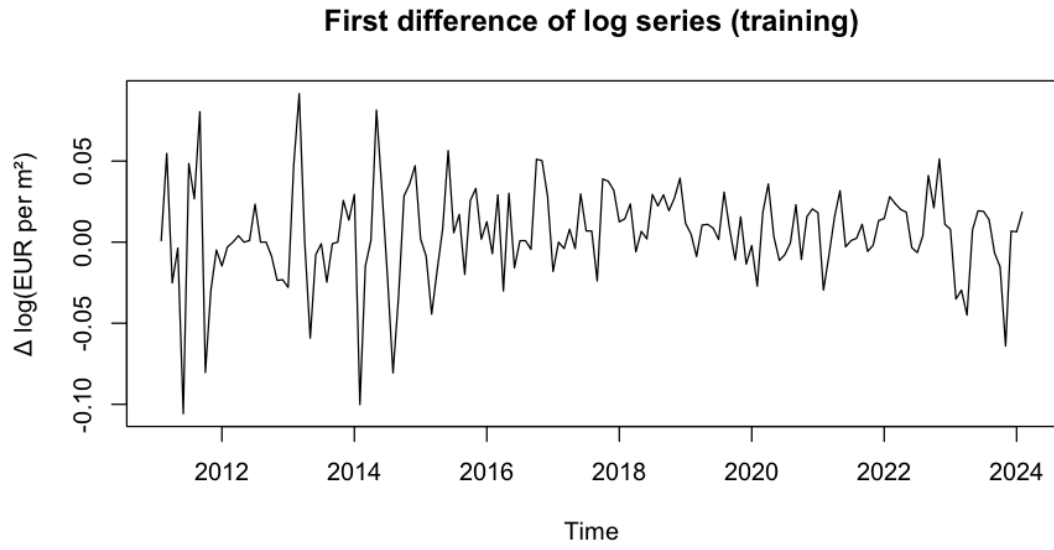


Figure 10: First difference of $\log(y_t)$ on the training sample.

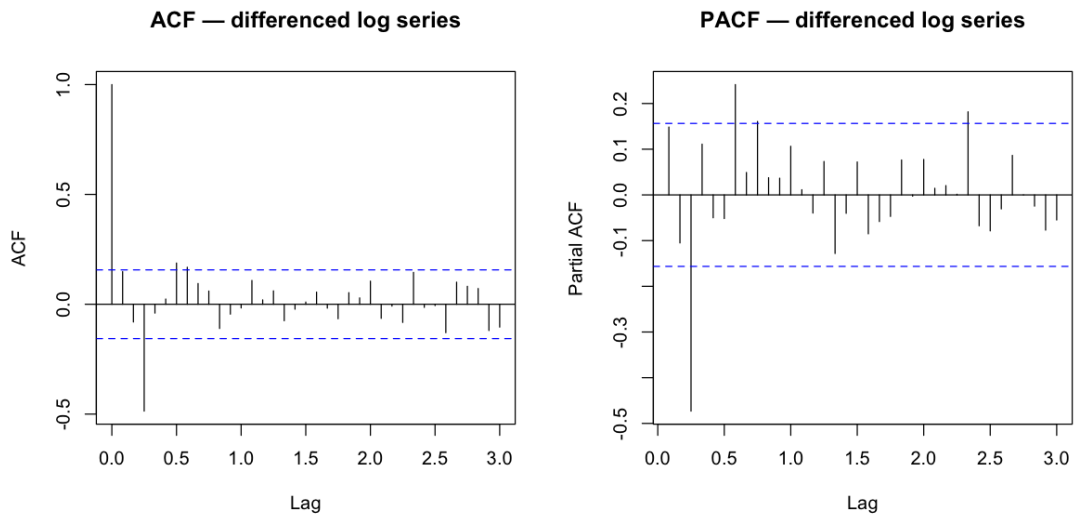


Figure 11: ACF and PACF of the differenced log series.

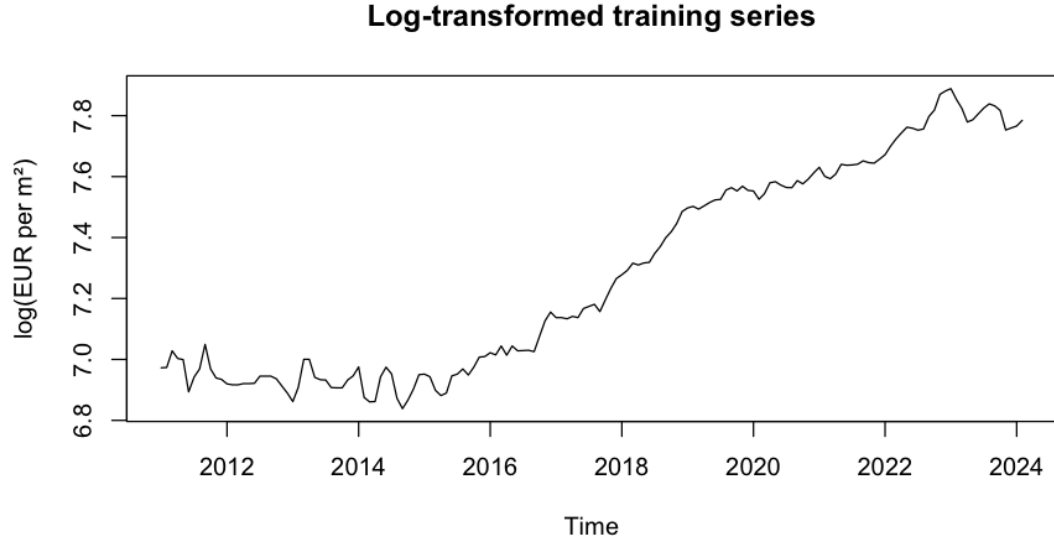


Figure 12: Log-transformed training series.

4.2 Model short-list and AIC

Three candidates are compared on the training sample (all with $d = 1$ on the log scale):

- **Candidate A:** `auto.arima` with `seasonal = FALSE`
- **Candidate B:** `auto.arima` with `seasonal = TRUE`
- **Candidate C:** manual SARIMA(3, 1, 0)(0, 0, 1)₁₂

Both auto-selected models yield **ARIMA**(1, 1, 4) ($\text{AIC} = -710.4$); the manual SARIMA has higher AIC ($= -693.4$). ARIMA(1, 1, 4) is retained for diagnostics and forecasting. Forecasts on the original scale are obtained by exponentiating the log-scale predictions.

4.3 Diagnostic checking

Figure 13 shows residual diagnostics. The Ljung-Box test on innovation residuals at lag 24 with `fitdf = 5` (estimated AR/MA coefficients) gives $Q^* = 16.80$, $\text{df} = 19$, $p = 0.60$, so residual autocorrelation is not significant at conventional levels.

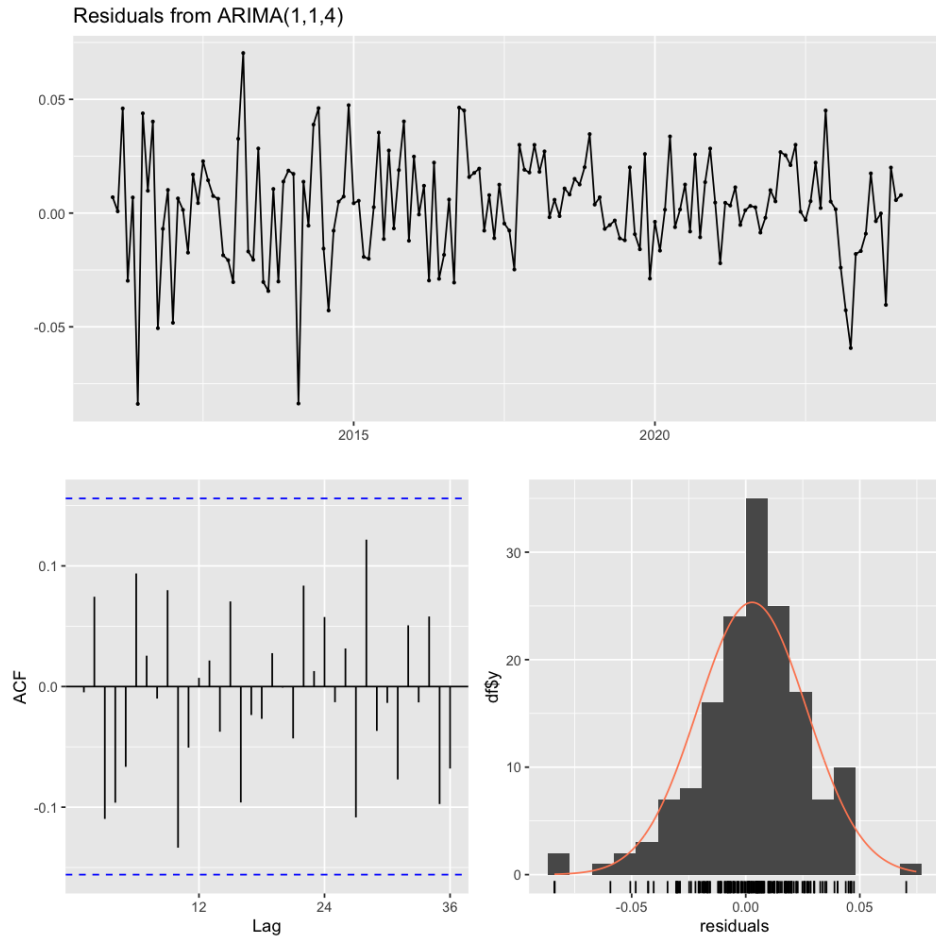


Figure 13: Residual diagnostics for ARIMA(1, 1, 4) on $\log(y_t)$.

Figure 14 compares the back-transformed forecast with the test set in EUR/m².

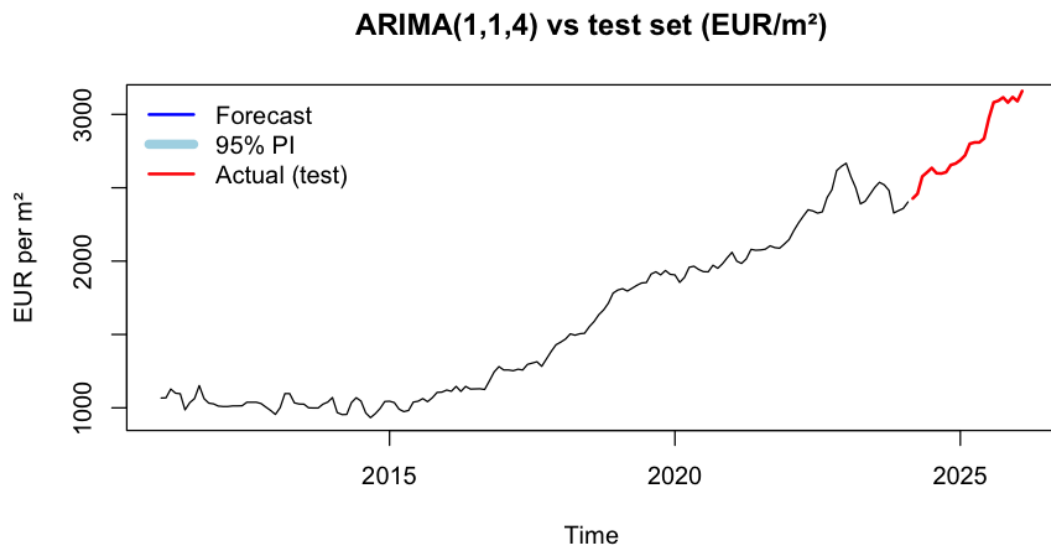


Figure 14: ARIMA(1, 1, 4) forecast vs. test set (EUR/m²).

5 Forecast comparison

Table 1 summarises test-set accuracy. **ETS** achieves the lowest RMSE (326 EUR/m²) and MAPE (9.6%). Holt-Winters (multiplicative) and STLTF perform similarly (RMSE $\approx$ 336). The damped Holt-Winters variant and the log-ARIMA model perform worse on this 24-month test window, with ARIMA showing the largest systematic under-forecast (MAPE 18.3%).

All methods produce positive mean error: actual prices in March 2024–February 2026 rise faster than any model extrapolates from the training history. The 95% intervals generally remain wide enough to contain much of the test path.

Table 1: Test-set accuracy (24-month hold-out).

Method	RMSE	MAE	MAPE (%)	ME
ETS	325.7	280.6	9.62	280.6
HW (mult.)	335.9	295.9	10.20	295.9
STLTF	336.3	294.5	10.13	294.5
HW (damped)	415.4	359.3	12.32	359.3
ARIMA(1, 1, 4) log	582.5	529.0	18.31	529.0

6 Out-of-sample forecasts

Each method is refitted on the full sample and used to forecast the next 24 months (March 2026–February 2028) with 95% prediction intervals (Figures 15–16). Full numerical output is in `outputs/tables/out_of_sample_forecasts_95.csv`.

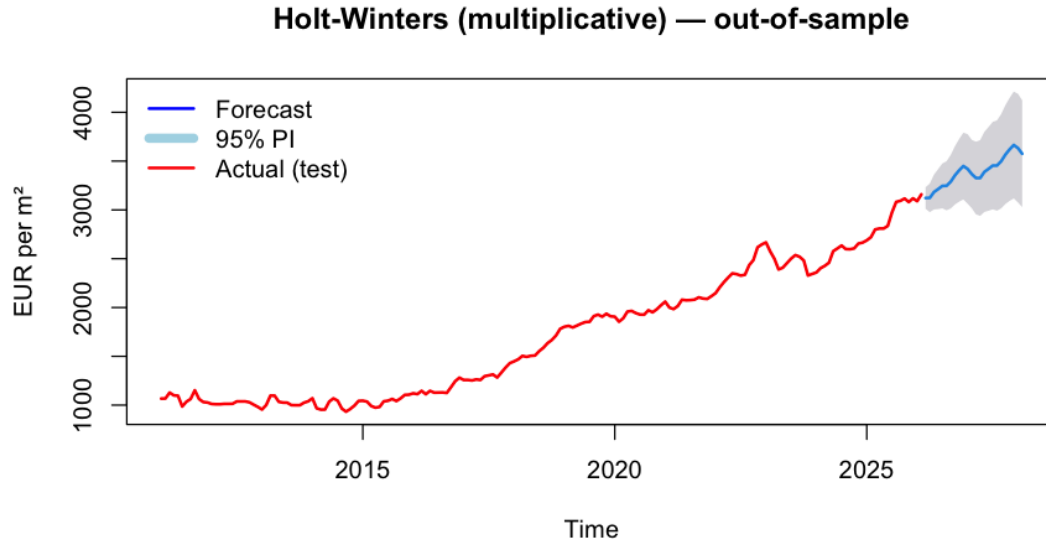


Figure 15: Holt-Winters (multiplicative) — out-of-sample forecast.

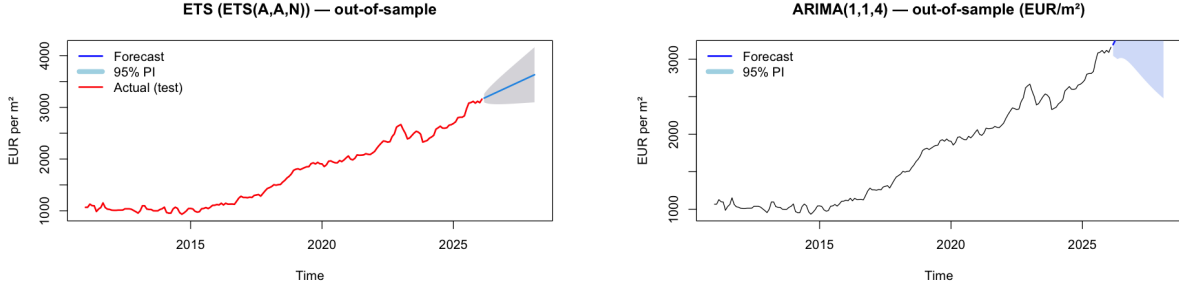


Figure 16: Out-of-sample forecasts: ETS (left) and ARIMA(1, 1, 4) (right).

7 Benefits and limitations

- **Multiplicative decomposition** respects proportional seasonality when the level trends strongly; for this series the seasonal effect is small but the choice is theoretically appropriate.
- **Holt-Winters (multiplicative)** captures trend and seasonality directly; performance is competitive though not best on the test set.
- **ETS** provides automatic model selection and achieves the lowest test RMSE here, but drops explicit seasonality on the training window.
- **STLF** combines robust STL with ETS; results are similar to Holt-Winters.
- **Log-ARIMA(1, 1, 4)** follows standard stationarity practice (log + differencing) and passes residual diagnostics, but under-performs on the test set when growth accelerates; back-transformed intervals assume log-normal innovations.
- **General limitation:** all methods assume historical patterns continue; none fully captures the 2024–2026 acceleration in prices.

8 Software and reproducibility

Analysis is implemented in R (`forecast`, `tidyverse`). The unified pipeline is `run_analysis.R` (sources `01_EDA.R`–`04_comparison_oos.R` and `R/utils.R`). Run `Rscript run_analysis.R` from the project root; figures and tables are written to `outputs/`. The PDF is built with `make` in `report/`.

References

- Hyndman, R.J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts.
- Portuguese housing price statistics (series 11A1312: Porto), `timeSeriesPorto.csv`.

Declaration of Authenticity

The authors declare that this work is original, that sources are properly cited, and that Generative AI tools were used to assist with code integration, LaTeX formatting, and pipeline structure. All numerical results and interpretations were verified against R output. AI systems are not listed as authors.

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